

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering
B. Tech (EE) | 5th Semester EE2 | Autonomy

Subject: Power Electronics Subject Code: EE-504

Teaching-Learning Method: Collaborative Learning (CL)

Date: Thursday, 3rd September 2026 Time: 3:00–3:50 PM Section: EE-2

Faculty: Kingsuk Majumdar, Ph.D. (Engg.) Dept. of EE, BCREC

i Topics: Group A (G01–G05): SCR Protection – Snubber Design, dv/dt and di/dt Protection, Gate Protection

Group B (G06–G10): Commutation of SCR – Natural Commutation vs Forced Commutation: Conditions, Circuit Analysis and Comparison

Module: 2 (PNPN Devices – Thyristors) **CO:** CO1, CO2 **Bloom's Level:** BL3 (Apply), BL4 (Analyse)

Reference: M. H. Rashid, *Power Electronics*, PHI, 3rd Ed., Ch. 4; P. S. Bimbhra, *Power Electronics*, Khanna Publishers, Ch. 3.

Instructions – Collaborative Learning:

- This is a **group activity**. Each group (4–5 students) works on its assigned problem set. Every member is expected to contribute.
- **Phase 1 – Group Discussion and Solution (30 min, 3:00–3:30 PM):** Discuss within your group. Each member writes their own solution on the individual response sheet. Submit one combined solution sheet per group.
- **Phase 2 – Presentation and Class Discussion (15 min, 3:30–3:45 PM):** One representative from each group presents the group's solution on the board (2–3 minutes per group). Other groups may question and respond.
- **Phase 3 – Faculty Consolidation (5 min, 3:45–3:50 PM):** Faculty corrects errors, emphasises key concepts, and connects to CT-1 scope.
- All sheets – individual and group – must be submitted before leaving. These are mandatory NBA documentation.
- Write **Group Number, Name, and Roll Number** on every sheet.

A Note on Coverage: Both topic sets (Protection and Commutation) are from Module 2 (SCR Thyristors). Lecture 2.4 covers SCR protection and series/parallel operation. Lecture 2.5 covers gate triggering circuits and commutation techniques. These topics are being covered in Week 4 (24-Aug-2026 onwards). The Collaborative Learning activity reinforces and extends the lecture content.

GROUP A (G01–G05): SCR Protection*Topics: Snubber Design | dv/dt Protection | di/dt Protection | Gate Protection***💡 Background (read before solving):**

An SCR can be damaged or triggered inadvertently by:

- **Excessive dv/dt :** A rapid rise of anode-cathode voltage charges the junction capacitance C_{J2} , producing a displacement current $i = C_{J2} dv/dt$ that triggers the SCR without a gate signal.
- **Excessive di/dt :** A rapid rise of anode current at turn-on creates localised heating near the gate region before current spreads across the entire junction area, potentially destroying the device.
- **Gate overvoltage:** Excessive gate voltage or reverse gate voltage damages the gate-cathode junction.

A **snubber circuit** (series R-C placed across the SCR) limits dv/dt . A series inductor limits di/dt at turn-on.

Problem A1 – Snubber Design**[4 marks]**

An SCR is connected in a resistive circuit with supply voltage $V_s = 200$ V. The maximum permissible dv/dt for the device is $(dv/dt)_{max} = 100$ V/ μ s. The load resistance is $R_L = 10$ Ω .

A series R-C snubber (R_s and C_s) is placed across the SCR.

- Write the expression for dv/dt across the snubber capacitor at $t = 0^+$ (when the SCR turns off and the circuit is modelled as a series R_L - R_s - C_s circuit with initial voltage V_s across the capacitor = 0). **[1 marks]**
- If $R_s = R_L = 10$ Ω and the peak dv/dt condition gives $C_s \geq V_s / [(R_L + R_s) (dv/dt)_{max}]$, calculate the minimum value of C_s . **[1 marks]**
- For the C_s found in (ii), calculate the snubber energy dissipated per switching cycle: $E_s = \frac{1}{2} C_s V_s^2$. **[1 marks]**
- If the circuit operates at a switching frequency of $f = 2$ kHz, calculate the average power dissipated in the snubber resistor. **[1 marks]**

Problem A2 – di/dt Protection and Gate Protection**[3 marks]**

- An SCR has a maximum di/dt rating of 50 A/ μ s. The supply voltage is $V_s = 300$ V. Calculate the minimum series inductance L_s required to limit di/dt at turn-on to the rated value. Use: $(di/dt)_{max} = V_s / L_s$. **[1 marks]**
- Explain with a neat circuit diagram the **gate protection** measures used for an SCR:
 - Resistance R_G in series with the gate to limit gate current
 - Resistance R_{GK} between gate and cathode to bypass leakage
 - Zener diode across the gate-cathode terminals to clamp voltage

For each, state what it protects against.

[2 marks]

- A Zener diode with breakdown voltage $V_Z = 10$ V is used for gate protection. The gate supply is 15 V and a series resistance $R_G = 100$ Ω is used. Calculate the gate current I_G when the Zener is conducting (i.e. the gate voltage is clamped at V_Z). **[1 marks]**

Problem A3 – Discussion Question**[3 marks]**

- i. Why is it necessary to use both a snubber (for dv/dt) AND a series inductor (for di/dt) in many practical SCR circuits? Can one replace the other? Justify your answer. **[1 marks]**
- ii. In a full-wave rectifier using SCRs, the snubber capacitor charges to V_s when the SCR is OFF. When the SCR fires, the capacitor discharges through R_s . Explain why R_s must not be too small (hint: di/dt through the SCR) and must not be too large (hint: dv/dt limiting effectiveness). **[2 marks]**

Group A Total: 10 marks**GROUP B (G06–G10): SCR Commutation**

Topics: Natural Commutation | Forced Commutation | Circuit Analysis | Comparative Study

💡 Background (read before solving):

Commutation is the process of turning OFF a conducting SCR. Since the gate cannot turn OFF the SCR once latched, the anode current must be reduced below the holding current I_H through circuit action.

Natural (Line) Commutation: The AC supply itself reverses the anode-cathode voltage periodically, naturally reducing I_A to zero. Applicable only in AC circuits (rectifiers, AC controllers).

Forced Commutation: An external circuit (L-C resonant circuit or auxiliary SCR) is used to force the anode current to zero. Used in DC circuits (choppers, inverters) where no natural voltage reversal is available.

Problem B1 – Natural Commutation Analysis**[4 marks]**

A single-phase full-wave controlled rectifier uses two SCRs (T1 and T2) with a purely resistive load $R = 20 \Omega$ fed from a 230 V, 50 Hz AC supply. The firing angle is $\alpha = 45^\circ$.

- i. Explain how natural commutation occurs in this circuit. Specifically, state: (a) which voltage drives I_A of T1 to zero, (b) at what angle does T1 turn off, and (c) why T1 cannot re-fire immediately after turn-off. **[2 marks]**
- ii. Calculate the average output voltage: $V_{dc} = \frac{V_m}{\pi}(1 + \cos \alpha)$ where $V_m = \sqrt{2} \times 230$ V. **[1 marks]**
- iii. The circuit turn-off time available for T1 is $t_q = (\pi - \alpha)/\omega$ (half-wave case) or $t_q = \alpha/\omega$ (full-wave case). For $\alpha = 45^\circ$ and $f = 50$ Hz, calculate t_q for the full-wave case. Verify that this exceeds the SCR turn-off time $t_{off} = 1.5$ ms. **[1 marks]**

Problem B2 – Forced Commutation Analysis**[3 marks]**

A Class A (self-commutated) forced commutation circuit has a commutating capacitor $C = 10 \mu\text{F}$ and commutating inductor $L = 100 \mu\text{H}$. The supply voltage is $V_s = 200$ V.

- i. Calculate the resonant frequency of the L-C commutation circuit: $f_r = \frac{1}{2\pi\sqrt{LC}}$. [1 marks]
- ii. The commutation time (time for which the main SCR sees reverse voltage) is approximately $t_c = \pi\sqrt{LC}$. Calculate t_c and verify it exceeds a typical $t_{off} = 20 \mu s$ for the SCR. [1 marks]
- iii. The peak current through the commutation circuit is $I_{peak} = V_s\sqrt{C/L}$. Calculate I_{peak} and explain its significance in the commutation process. [1 marks]

Problem B3 – Comparison and Analysis**[3 marks]**

- i. Complete the following comparison table:

Parameter	Natural Commutation	Forced Commutation
Supply type required		
External commutation circuit		
Applicable converter types		
Circuit complexity		
Power factor		

[1 marks]

- ii. A DC chopper (step-down) operates from a 100 V DC supply and feeds a resistive load. The SCR used has $t_{off} = 25 \mu s$.
 - (a) Can natural commutation be used here? Give a clear reason.
 - (b) What type of forced commutation would you recommend? Name one specific class (Class A through Class F) and explain the principle in one sentence.

[2 marks]**Group B Total: 10 marks**

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering

CO-PO Mapping of Collaborative Learning Activity

Subject: Power Electronics (EE-504) Section: EE-2 Date: 3rd September 2026

1 Course Outcomes Addressed

CO1: Define and explain the characteristics and operation of power electronic devices and switching components.

CO2: Analyze and design different types of power electronic converters and their control circuits.

Bloom's Level: BL3 (Apply) for numerical problems A1, A2(i), B1(ii), B1(iii), B2(i)–(iii) | BL4 (Analyse) for discussion problems A2(ii), A3, B3

Topics: SCR Protection (Snubber, dv/dt , di/dt , Gate) | Natural Commutation | Forced Commutation | Comparative Analysis

References: M. H. Rashid, *Power Electronics*, PHI, 3rd Ed., Ch. 4; P. S. Bimbhra, *Power Electronics*, Khanna Publishers, Ch. 3.

2 CO-PO Mapping Table

CO	Course Outcome	Programme Outcomes (PO)											PSO		
		1	2	3	4	5	6	7	8	9	10	11	1	2	3
CO1	Define and explain characteristics and operation of power electronic devices	3	2	1	2	2	1	1	–	1	1	2	3	2	1
CO2	Analyze and design power electronic converters and control circuits	3	3	3	2	3	1	1	1	2	2	2	3	3	1

Table 1: CO-PO/PSO Correlation (3: High, 2: Medium, 1: Low, –: No correlation), as per EE-504 lesson plan

3 Problem-wise CO-PO Justification

Problem	Skill / Concept Tested	CO	PO	BL
A1(i,ii)	Apply snubber design formula; compute minimum C_s to meet dv/dt rating	CO2	PO1, PO2	BL3
A1(iii,iv)	Compute snubber energy loss and average power dissipation at given switching frequency	CO2	PO1, PO2	BL3
A2(i)	Apply di/dt constraint to find minimum series inductance	CO2	PO1, PO2	BL3
A2(ii)	Analyse and explain gate protection circuit elements and their individual functions	CO1	PO1, PO2	BL4
A2(iii)	Apply Zener clamp circuit to compute gate current	CO1	PO1, PO2	BL3
A3(i)	Analyse why dv/dt and di/dt protections serve different purposes and cannot be mutually replaced	CO1	PO1, PO2	BL4
A3(ii)	Analyse snubber resistor sizing trade-off between di/dt and dv/dt limiting effectiveness	CO2	PO1, PO2, PO3	BL4
B1(i)	Analyse the natural commutation mechanism: voltage reversal, extinction angle, and SCR blocking	CO1	PO1, PO2	BL4
B1(ii,iii)	Apply rectifier output voltage formula; compute circuit turn-off time and verify adequacy	CO2	PO1, PO2	BL3
B2(i)–(iii)	Apply resonant L-C circuit formulas to compute f_r , commutation time t_c and peak current	CO2	PO1, PO2	BL3
B3(i)	Analyse and compare natural vs forced commutation on five parameters	CO1, CO2	PO1, PO2	BL4
B3(ii)	Analyse applicability of commutation methods to a DC chopper; justify forced commutation selection	CO2	PO1, PO2, PO3	BL4

Table 2: Problem-wise skill, CO, PO and Bloom's Level mapping

4 PO Description Reference

PO	Description (AICTE/NBA)
PO1	Engineering Knowledge: Apply mathematics, science, and engineering fundamentals to solve complex power electronics problems
PO2	Problem Analysis: Identify, formulate, and analyse complex engineering problems using device and circuit equations
PO3	Design/Development of Solutions: Evaluate design choices (snubber component values, commutation circuit topology) against performance requirements

Table 3: Relevant PO descriptions for CO1 and CO2 mapping

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering

Faculty Evaluation Report

Teaching-Learning Method: Collaborative Learning (CL)

Subject: Power Electronics (EE-504) Section: EE-2

Date: Thursday, 3rd September 2026 Time: 3:00–3:50 PM

Venue: _____ Faculty: Kingsuk Majumdar, Ph.D. (Engg.)

5 Activity Details

Activity	Collaborative Learning (CL)
Subject / Code	Power Electronics / EE-504
Section	EE-2 (EE2X + EE2Y), B. Tech (EE), 5th Semester
Batch	2024–28
Session	2026–27 (ODD Semester)
Date and Time	Thursday, 3rd September 2026, 3:00–3:50 PM
Venue	_____
Total Students Present	_____ out of 47
Groups	10 groups (G01–G10); 4–5 students per group
Group A (G01–G05)	SCR Protection – Snubber, dv/dt , di/dt and Gate Protection
Group B (G06–G10)	SCR Commutation – Natural and Forced Commutation: Analysis and Comparison
Module	Module 2 – PNP Devices (Thyristors)
CO Addressed	CO1 (Define and explain SCR device characteristics); CO2 (Analyse power electronic circuits)
Bloom's Level	BL3 (Apply) for numericals BL4 (Analyse) for discussion and comparison

Table 4: Collaborative Learning Activity Details – EE-504

6 🕒 Phase-wise Time Observation

Phase	Duration	Time Slot	Faculty Observation
Group Discussion	30 minutes	3:00–3:30 PM	_____
Presentation	15 minutes	3:30–3:45 PM	_____
Consolidation	5 minutes	3:45–3:50 PM	Faculty correction and summary on board

Table 5: Phase-wise time and observation log

7 ⚠️ Common Mistakes to Watch For

Fill after conducting the activity. Typical mistakes:

- A1 – C_s formula inversion:** Students write $C_s = (dv/dt)_{max} \times (R_L + R_s)/V_s$ instead of $V_s/[(R_L + R_s) \times (dv/dt)_{max}]$. The units check: $[V/((\Omega) \times (V/s))] = [s/\Omega] = [F]$. _____ (observed: Yes / No)
- A1(iv) – Power formula error:** Students compute $P_s = E_s/f$ (divide) instead of $P_s = E_s \times f$ (multiply). Power increases with frequency. _____ (observed: Yes / No)
- A2(i) – di/dt formula:** Students use $L_s = (di/dt) \times V_s$ instead of $L_s = V_s/(di/dt)$. _____ (observed: Yes / No)
- B1(iii) – Turn-off time formula:** Students confuse half-wave and full-wave expressions. For full-wave: $t_q = \alpha/\omega$, not $(\pi - \alpha)/\omega$. _____ (observed: Yes / No)
- B2 – $\sqrt{LC}$ unit error:** Students do not resolve the units of $\sqrt{LC}$ correctly. L in H, C in F: $\sqrt{LC}$ is in seconds. $\sqrt{100 \times 10^{-6} \times 10 \times 10^{-6}} = 31.62 \mu s$. _____ (observed: Yes / No)
- B3 – Natural commutation in DC circuit:** Students state that natural commutation is possible in a DC chopper because “the supply can be reversed”. This is incorrect. Natural commutation requires a naturally alternating supply voltage. DC choppers always need forced commutation. _____ (observed: Yes / No)
- Additional mistakes observed:**

8 Group Performance Summary

Group	Members (Roll Nos.)	Topic	Presenter	Score / 10
G01	_____	Protection (A)		
G02	_____	Protection (A)		
G03	_____	Protection (A)		
G04	_____	Protection (A)		
G05	_____	Protection (A)		
G06	_____	Commutation (B)		
G07	_____	Commutation (B)		
G08	_____	Commutation (B)		
G09	_____	Commutation (B)		
G10	_____	Commutation (B)		

Table 6: Group performance table (fill after evaluation)

9 ★ Key Learning Points Emphasised During Consolidation

1. The snubber circuit limits dv/dt – it does NOT limit di/dt . A separate series inductor is needed for di/dt protection. These are two distinct failure mechanisms requiring two distinct protective elements.
2. The snubber resistor R_s plays a dual role: it damps the oscillation between C_s and L_{stray} and limits the discharge current of C_s at SCR turn-on. Too small: large di/dt at discharge; too large: dv/dt limiting is ineffective.
3. Natural commutation is inherently simple and efficient because the supply does the work. Forced commutation requires additional L-C components, increases circuit complexity and cost, and the commutating capacitor must always be pre-charged. Modern SCR circuits in DC applications increasingly use IGBTs which can be gate-turned-off.
4. The commutation time $t_c = \pi\sqrt{LC}$ must always exceed the SCR turn-off time t_{off} . If $t_c < t_{off}$, the SCR re-fires before it can properly recover – commutation fails.
5. Additional points noted:

10 Faculty Remarks and Overall Assessment

Overall quality of group discussions:

Quality of board presentations:

Cross-group interaction and questioning:

Effectiveness of Collaborative Learning for these topics:

Action to be taken before CT-1 (09-Sep-2026):

Kingsuk Majumdar, Ph.D. (Engg.)

Assistant Professor, Dept. of EE
Dr. B. C. Roy Engineering College, Durgapur

Date: _____

Signature: _____

11 NBA 2 TLM – Collaborative Learning Activity (EE-504)

11.1 Activity Overview

TLM Method:	Collaborative Learning (CL)
Subject / Code:	Power Electronics / EE-504
Section:	EE-2 (EE2X + EE2Y), B. Tech (EE), 5th Semester, Batch 2024–28
Date and Time:	Thursday, 3rd September 2026, 3:00–3:50 PM
Venue:	
Faculty:	Kingsuk Majumdar, Ph.D. (Engg.), Dept. of EE, BCREC
Group A Topics:	SCR Protection – Snubber Design (dv/dt), di/dt Protection (series inductor), Gate Protection
Group B Topics:	Commutation of SCR – Natural (Line) Commutation and Forced Commutation: conditions, circuit analysis, comparison
Module:	Module 2 – PNP Devices (Thyristors)
CO Addressed:	CO1 (Define and explain SCR device operation); CO2 (Analyse power electronic circuits)
Bloom's Level:	BL3 (Apply) for numerical problems; BL4 (Analyse) for discussion and comparison problems
Total Students:	47 (EE2X: 24 students, EE2Y: 23 students)
Groups:	10 groups of 4–5 students each

11.2 Group Allotment – EE-504 CL Activity

Group	Topic Set	Roll Numbers	Members
G01	Protection (A)	120016244052, 120016244053, 120016244054, 120016244055	_____
G02	Protection (A)	120016244056, 120016244057, 120016244058, 120016244059	_____
G03	Protection (A)	120016244060, 120016244061, 120016244062, 120016244064	_____
G04	Protection (A)	120016244065, 120016244066, 120016244067, 120016253001	_____
G05	Protection (A)	120016253002, 120016253003, 120016253004, 120016253005, 120016253006	_____
G06	Commutation (B)	120016253007, 120016253008, 120016253009, 120016253010	_____
G07	Commutation (B)	120016253011, 120016253012, 120016253013, 120016253014, 120016253015	_____
G08	Commutation (B)	120016253016, 120016253017, 120016253018, 120016253019, 120016253020	_____
G09	Commutation (B)	120016253021, 120016253022, 120016253023, 120016253024, 120016253025	_____

Group	Topic Set	Roll Numbers	Members
G10	Commutation (B)	120016253026, 120016253028, 120016253029, 120016253030, 120016253031, 120016253032, 120016253033	_____

i Note: Total 47 students. Groups G01–G09 have 4–5 students each. G10 has 7 students (EE2Y tail group) to exhaust the roll list. Roll 120016244063 is absent from the register. G05 has 5 members; G07, G08, G09 have 5 members each.

11.3 Rationale for Topic Selection and Group Format

SCR protection and commutation are complementary aspects of Module 2 that extend beyond the basic V-I characteristics and two-transistor model covered in Lectures 2.1–2.3. Assigning protection to one cluster of groups (G01–G05) and commutation to another (G06–G10) allows each group to develop deeper expertise in one area, while the cross-group presentation phase transfers knowledge across both sets. Protection involves primarily design calculations (snubber sizing, inductance calculation) that are well-suited to collaborative numerical problem solving in 30 minutes. Commutation involves analytical derivations (resonant frequency, commutation time) and conceptual comparison that benefit from peer discussion – students who grasp the circuit mechanism quickly can explain it to group members who are still building intuition. The activity also directly reinforces CT-1 scope (Modules 1, 4, and 2 scheduled 09–September-2026), making it serve simultaneously as a structured revision session.

11.4 Phase-wise Execution Plan

Phase	Duration	Time	Description
Group Discussion	30 min	3:00–3:30 PM	Groups G01–G05 solve Protection problems (A1–A3); Groups G06–G10 solve Commutation problems (B1–B3). Faculty circulates between groups, notes engagement, provides one clarifying hint per group maximum.
Presentation	15 min	3:30–3:45 PM	One representative from each group presents on the board (approx. 90 seconds per group for 10 groups). G01 presents A1; G06 presents B1; others fill in remaining parts. Cross-group questions encouraged.
Consolidation	5 min	3:45–3:50 PM	Faculty corrects key errors (especially C_s formula, t_c vs t_{off} condition, and natural commutation inapplicability in DC circuits). Connects to CT-1 revision and Remedial class scope.

Table 8: Phase-wise execution plan for Collaborative Learning – EE-504

11.5 CO-PO Mapping Summary

CO	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	1	3	2	1
CO2	3	3	3	3	3	1
3 = High, 2 = Medium, 1 = Low, – = No correlation						

Table 9: CO-PO-PSO summary for Collaborative Learning activity

11.6 Supporting Documents for NBA File Submission

Sl.	Document	Format
1	Group allotment list with Name, Roll No., Group No. (Table in Section 3.2 above)	PDF
2	CL Problem Sheet – Group A (Document 1, pages 1–3)	PDF
3	CL Problem Sheet – Group B (Document 1, pages 3–5)	PDF
4	Individual response sheets – minimum 10 samples (one per student, across both topic sets)	Single PDF
5	Group combined solution sheets – all 10 groups	Single PDF
6	Score sheet (Table 3 of Faculty Evaluation Report, filled)	PDF
7	Photographs: groups working and presenting (GEO tag + non-GEO tag)	Single PDF
8	Faculty Evaluation Report (Document 3)	PDF
9	CO-PO mapping of activity (Document 2)	PDF

Table 10: Documents to be submitted for CL NBA 2 file – EE-504

Document Prepared by: Kingsuk Majumdar, Ph.D. (Engg.)

Department of Electrical Engineering
Dr. B. C. Roy Engineering College, Durgapur

Date: 30-August-2026 **Version:** V3 (breakable added)

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